

Project 1: Toxic Comment Classification

Objective

Your task is to build models to detect toxic comments. This project extends what we did in class with IMDb sentiment analysis, but the data and challenges are different:

- The dataset is more imbalanced (fewer toxic comments than non-toxic).
- Toxicity is harder to capture.

Grading

You will deliver:

1. **Report:** PDF, max 2 pages; answers to questions with figures/tables
2. **Notebook:** .py; should be reproducible with a fixed random state
3. **10 minutes oral presentation**

Use of external libraries and LLMs (e.g. ChatGPT):

- State any AI/tool assistance you used and how

Dataset

We will use a labelled dataset. Each observation is a comment from Wikipedia with a binary label: toxic (1) or not toxic (0).

Tasks

1. **Preprocessing.** Apply the same steps as in class: tokenization, stop-words, cleanup, stemming or lemmatization.
 - Which preprocessing step would you skip here, and why?
 - Which stopword would you keep in this task, and why?
 - Propose one preprocessing tweak that might help toxic detection and that we have not covered.
2. **Feature engineering.** Use at least two bag of words representation with different N-grams. Which representation is most useful here? Why?
3. **Modeling.**
 - Train at least three machine learning models: Logistic Regression + Linear SVM + one model of your choice
 - Build a simple baseline using a dictionary of toxic cue words. Justify how you chose these words. Predict “toxic” if one of these words appears.
4. **Evaluation.** For each model:
 - Produce a classification report (precision, recall, F1)
 - Calculate the area under the curve
5. **Reflection.** Suppose Wikipedia deploys your model.
 - Which metric should matter most: accuracy, precision, or recall?
 - Discuss the consequences of each error type (False positive and false negative). Who is harmed in each case (the commenter, other users, moderators, the platform?)
 - Consider context: would your answer change if the model was used for automatic blocking versus as a moderator support tool? Why?
 - Compare this trade-off to a real-world analogy such as COVID testing. How is moderation on Wikipedia similar or different?